hybrid inheritance in c++ explain in simple and eays way along with example

Hybrid Inheritance in C++ (Simple Explanation)

Hybrid inheritance is a combination of **two or more types of inheritance** in one program.
It usually combines **multiple inheritance** with **multilevel inheritance**.

👉 In simple words:
When a class is derived using **more than one inheritance type together**, it is called **Hybrid Inheritance**.

Think of it like a **family tree where different inheritance patterns mix together**.

Basic Idea Diagram

```
    A
   / \
  B   C
   \ /
    D
```

- **B and C inherit from A** (Hierarchical inheritance)
- **D inherits from B and C** (Multiple inheritance)

Together this forms **Hybrid inheritance**.

Example in C++

```cpp
#include <iostream>
using namespace std;
```

```cpp
class A {
public:
    void showA() {
        cout << "Class A function" << endl;
    }
};

class B : public A {   // B inherits A
public:
    void showB() {
        cout << "Class B function" << endl;
    }
};

class C : public A {   // C also inherits A
public:
    void showC() {
        cout << "Class C function" << endl;
    }
};

class D : public B, public C {   // D inherits B and C
public:
    void showD() {
        cout << "Class D function" << endl;
    }
};

int main() {
    D obj;

    obj.showB();
    obj.showC();
    obj.showD();
```

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Output

```
Class B function
Class C function
Class D function
```

Key Points (Very Easy)

- Hybrid inheritance = **Combination of different inheritance types**
- Mostly combines:
  - **Hierarchical inheritance**
  - **Multiple inheritance**
- It creates a **diamond-like structure**

Real-Life Example

Imagine:

- **A = Person**
- **B = Student**
- **C = Employee**
- **D = WorkingStudent**

```
Person
  /  \
Student Employee
     \  /
   WorkingStudent
```

A **WorkingStudent** is both a **Student** and an **Employee**, which comes from **Person**.

✅ **One-line definition for exams:**

> Hybrid inheritance is a type of inheritance that combines more than one inheritance type (such as multiple and hierarchical inheritance) in a single program.

If you want, I can also show:

- **Hybrid inheritance diamond problem** (very important for exams)
- **Hybrid inheritance with virtual base class** (commonly asked in viva) 📚

multiple and multileel and single and all other types of inheritance in cpp

In **C++**, inheritance allows a class (**child/derived class**) to get properties and functions from another class (**parent/base class**).

There are **5 main types of inheritance in C++**.

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1. Single Inheritance

👉 **One child class inherits from one parent class.**

Structure

```
A → B
```

Example

```cpp
#include <iostream>
using namespace std;

class A {
public:
    void display() {
        cout << "This is class A" << endl;
    }
};

class B : public A {
};

int main() {
    B obj;
```

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✅ **Output**

```
This is class A
```

---

2. Multilevel Inheritance

👉 A class inherits from another class which is already inherited from another class.

Structure

```
A → B → C
```

Example

```
#include <iostream>
using namespace std;

class A {
public:
    void showA() {
        cout << "Class A" << endl;
    }
};

class B : public A {
public:
    void showB() {
        cout << "Class B" << endl;
    }
};

class C : public B {
public:
    void showC() {
        cout << "Class C" << endl;
    }
};

int main() {
    C obj;
    obj.showA();
```

```
    obj.showB();
```

C++

---

3. Multiple Inheritance

👉 A class inherits from **more than one parent class**.

Structure

```
A   B
 \ /
  C
```

Example

```
#include <iostream>
using namespace std;

class A {
public:
    void showA() {
        cout << "Class A" << endl;
    }
};

class B {
public:
    void showB() {
        cout << "Class B" << endl;
    }
};

class C : public A, public B {
};

int main() {
    C obj;
    obj.showA();
```

C++

4. Hierarchical Inheritance

👉 Multiple child classes inherit from one parent class.

Structure

```
   A
  / \
 B   C
```

Example

```cpp
#include <iostream>
using namespace std;

class A {
public:
    void showA() {
        cout << "Class A" << endl;
    }
};

class B : public A {
};

class C : public A {
};

int main() {
    B obj1;
    C obj2;

    obj1.showA();
```

‹/› C++

5. Hybrid Inheritance

👉 Combination of two or more inheritance types.

Structure

```
    A
   / \
  B   C
   \ /
    D
```

Example

```cpp
#include <iostream>
using namespace std;

class A {
public:
    void showA() {
        cout << "Class A" << endl;
    }
};

class B : public A {
};

class C : public A {
};

class D : public B, public C {
};

int main() {
    D obj;
```

‹/› C++

Quick Summary Table

Type	Description
Single	One parent → one child
Multilevel	Parent → child → grandchild

Type	Description
Hierarchical	One parent → multiple children
Hybrid	Combination of different inheritance types

✅ Very Short Exam Definition

- **Single**: One base class and one derived class.
- **Multilevel**: A class derived from another derived class.
- **Multiple**: A class derived from more than one base class.
- **Hierarchical**: Multiple classes derived from one base class.
- **Hybrid**: Combination of two or more inheritance types.

If you want, I can also show a **very easy diagram + 5-line exam definitions (perfect for writing in paper)** 📄.